TinkerWithMe

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AI for Social Good · Full Curriculum

3-session series (90 min each) · 12–16 yrs · Intermediate → Advanced

Full facilitator curriculum PREMIUM

Overview

Solve a real community problem with AI, ending with a pitch and a Changemaker badge.

DETAIL	
Track	AI Skills for Kids
Format	3-session series (90 min each)
Ages	12–16 yrs
Level	Intermediate → Advanced
Price	from KES 8,000
Standalone	Yes
Prerequisites	None — standalone

Course Overview

AI for Social Good is a three-session studio in which small teams of students choose a real problem from their own community — water access, matatu routes, waste management, or school safety — and use artificial intelligence tools to research it, prototype a response to it, and pitch that response to a live audience of peers and parents. The course treats AI not as a toy but as a research assistant, a builder, and a communications partner, while keeping the human problem — and the people affected by it — at the centre of every decision.

The arc moves deliberately from head to hands to voice. Session 1 (Research) grounds students in their chosen challenge through AI-assisted research, mapping, and real interview practice, so that nothing gets built until the problem is well understood. Session 2 (Prototype) turns that understanding into a working first version of an AI-assisted solution — a chatbot, an image classifier, a data dashboard, or an AI-generated awareness campaign — using fast, free, no-code tools that a 12–16 year old can master in under an hour. Session 3 (Present) is a public showcase: teams pitch to a judging panel of facilitators and parents using a Problem–Solution–Impact–Ask structure, receive structured feedback, and one team is awarded the "TinkerWith_ Changemaker" badge.

Every student leaves with a completed project portfolio (problem brief, working prototype link/export, pitch deck) and a Changemaker certificate, whether or not their team wins the badge. The course is designed so a facilitator can run it end-to-end from this document with no additional prep beyond printing handouts and confirming Wi-Fi and accounts.

Learning Outcomes

- Scope a broad social problem down into a specific, research-backed, solvable challenge statement.
- Use generative AI (Gemini/ChatGPT) responsibly for research, including recognising when an AI answer needs human verification.
- Design and conduct open-ended interviews/surveys and translate findings into a visual problem map.
- Build a working AI-assisted prototype using at least one no-code AI tool (Google AI Studio, Teachable Machine, Google Sheets, or Canva Magic Media).
- Apply a define–build–test–iterate loop, incorporating peer feedback into a revised prototype.
- Structure and deliver a confident, evidence-based public pitch, and give constructive feedback to others.
- Reflect critically on the strengths and limits of AI tools when applied to real community problems.

Materials Master List

ITEM	SPEC	QTY PER STUDENT/ GROUP	NOTES
			Needed in all 3 sessions; must be able to reach Google,

Laptop or Chromebook with webcam	4GB+ RAM, updated Chrome browser, working camera and microphone	1 per 2 students	Canva and Teachable Machine sites
Reliable Wi-Fi / router	Min. 20 Mbps shared bandwidth	1 per venue	Critical — all build tools are browser-based. Have a mobile hotspot as backup (Safaricom/ Airtel)
Google account (student or shared class login)	Gmail-based; facilitator-managed shared login for students under 13 with parent consent obtained beforehand	1 per team of 3–4	Needed for Gemini/AI Studio, Forms, Sheets, My Maps
Canva for Education account	Free education tier, teacher-created class code	1 per team	Used for problem briefs, posters and Magic Media AI images
Printed Research Packet	Challenge cards, interview question template, problem-brief template (A4, stapled)	1 per student	Session 1
Printed local area map	A3, local estate/neighbourhood, with grid reference	1 per team	Session 1 mapping backup if internet drops
Sticky notes, assorted colours	3x3 in pads	2 pads per team	Brainstorming and dot-voting
Marker pens, fine + broad tip	Assorted colours	1 set per team	Available at Nerokas or Pixel Electronics stationery counters
Poster board / flip-chart paper	A1 size	1 per team	Problem brief display and pitch visual aid
Smartphone with camera	Any working camera	1 per team	Photographing training images and interview subjects (with consent)
Power strips / extension cables	4–6 outlet, surge-protected	3–4 per venue	Available at Nerokas or Pixel Electronics
Timer / visible countdown	Phone timer projected, or physical stopwatch	1 per venue	Pitch timing in Session 3
Projector and screen	HDMI-compatible, with audio out	1 per venue	Demoing tools and team slide decks
Name tags / lanyards	—	1 per student	Helps judges and parents in Session 3
			Session 3

Judging score sheets, printed	Rubric-based (see Assessment section)	1 per judge per team	
Changemaker certificates, printed and personalised	A5/A4 cardstock, pre-designed template	1 per student	Take-home; print before Session 3
Changemaker badge	Physical pin or sticker set, "TinkerWith_ Changemaker" design	1 set for winning team	Award item for the showcase
Portfolio folder	Physical folder or shared Google Drive folder template	1 per student	Holds problem brief, prototype link/screenshots, pitch deck
Snacks/refreshments	Light snacks and drinks	As needed	For the parent showcase in Session 3
Peer feedback forms	Printed half-sheet or Google Form	1 per team per testing round	Session 2 prototype testing

Session Plans

Session 1: Research — Pick a Local Challenge and Map It

Objectives

- Narrow a broad community issue into a specific, researchable challenge statement.
- Use AI research prompts to gather background facts and precedent solutions, and flag claims that need human verification.
- Design an interview/survey and produce a visual "problem map" combining data and location.

Timing (90 min)

- 0–10 min: Welcome, hook story (a short local news clip or facilitator anecdote about matatu delays, a broken tap, or a road accident near a school), introduce the four challenge areas.
- 10–25 min: Sticky-note brainstorm of specific local problems within the four areas; dot-vote; form teams of 3–4 around a chosen, narrowed challenge.
- 25–45 min: AI-assisted research sprint using Gemini or ChatGPT with guided prompts.

- 45–60 min: Draft and refine 5 interview/survey questions with AI help; practice-interview in pairs.
- 60–75 min: Build the problem map (Google My Maps or paper map) using research and interview data.
- 75–85 min: Teams complete a one-page Problem Brief (Canva or paper template).
- 85–90 min: Two teams share their brief; preview Session 2 build tracks.

Step-by-step

1. Open with the hook story and ask: "Who here has experienced something like this in the last month?" Collect 2–3 raised-hand stories.
2. Introduce the four challenge areas on the board: water access, matatu routes, waste management, school safety. Hand out sticky notes; each student writes 2 specific local problems (e.g., "broken tap at Kawangware Stage A," not "water scarcity in Kenya").
3. Cluster sticky notes on the board by theme. Facilitator leads a 5-minute dot-vote (3 dots per student) to surface the most energising, specific problems.
4. Form teams of 3–4 around the top-voted, sufficiently narrow problems. If a problem is still too broad, apply the "5 Whys" technique live with that team to narrow it (e.g., "Water access is a problem" → "why?" → repeat 5 times until you reach a specific tap, route, or school).
5. Hand out the Research Packet. Walk the whole room through the AI research workflow once on the projector (see Build detail), then let teams repeat it on their own machines.
6. Circulate as teams research; ask each team to name one AI claim they will verify with a human source (parent, shopkeeper, teacher) before Session 3.
7. Teach the interview question workshop: model one leading question ("Don't you think water is a big problem here?") versus one open question ("What happens on the days there's no water?"). Teams draft 5 questions and refine them with AI, then practice-interview each other in pairs for 2 minutes each.
8. Introduce Google My Maps (or the paper map backup). Demonstrate adding one pin with a description and colour-coding by severity. Teams then plot 4–6 locations relevant to their challenge.
9. Teams complete the Problem Brief template: Challenge statement, 3 causes, 2 precedent solutions, 5 interview questions, map screenshot/photo.

10. Two volunteer teams share their brief aloud; facilitator gives one strength and one sharpening question to each.

Build detail

Tools: Google Gemini (gemini.google.com) or ChatGPT (chat.openai.com) for research; Google My Maps (mymaps.google.com) for mapping; Canva or paper template for the Problem Brief.

Setup: Confirm each team can log in to a Google account before the session starts. Open a blank Gemini chat and a blank My Maps map on the projector as the live demo model.

Workflow:

1. In Gemini/ChatGPT, type the research prompt below, replacing the bracketed challenge with the team's actual topic.
2. Read the AI's answer together; underline any statistic or claim; ask "how would we check this?" and write the verification method in the Problem Brief.
3. Ask the follow-up prompt to generate interview questions, then edit at least 2 of them as a team to make them more specific to the local context.
4. Open My Maps, click "Add layer," then use the pin tool to drop markers at each location mentioned in research or interviews; colour pins red (severe), yellow (moderate), green (working well).

Example prompts:

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Prompt 1 (research):
"I'm a student researching [broken water taps] in [Kawangware, Nairobi]. Give me:
1) 3 likely root causes of this problem locally
2) 2 examples of how communities elsewhere have used technology or AI to address a similar problem
3) 3 statistics or data points I could try to verify, and what kind of source I should look for (e.g. NGO
report, county government data, news article)
Keep the whole answer under 200 words and use simple language."

Prompt 2 (interview questions):
"Turn the causes above into 5 open-ended, non-leading interview questions I could ask a local shopkeeper
or matatu driver. Avoid yes/no questions. Write them in simple English."
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What a good result looks like: a short, plain-language problem summary with 3 causes and 2 tech precedents, a list of 5 open-ended interview questions the team has personally edited, and a map with at least 4 colour-coded pins tied to real locations or data points.

Test/iterate: teams swap Problem Briefs with another team and check against a 3-item checklist: Is the problem specific to one place? Is at least one AI claim flagged for human verification? Are the interview questions open-ended? Teams revise before moving on.

Checks for understanding

- Why did we narrow our problem to one street, school, or stage instead of "all of Nairobi"?
- What is one thing the AI told us that we still need to check with a real person or source, and how will we check it?
- What makes the difference between a leading question and an open-ended one?

Common mistakes & fixes

MISTAKE	FIX
Problem stays too broad ("waste in Kenya")	Run the "5 Whys" live with the team until they land on a specific site, street, or institution
Team accepts AI statistics uncritically	Require every team to name and later check one AI claim against a human or local source
Interview questions are leading or closed	Rewrite using "what," "how," or "why" starters; ban yes/no phrasing
Team can't agree on a challenge	Facilitator runs a timed (5 minute) dot-vote and moves on; remind them the topic can be refined next session

Session 2: Prototype — Build an AI-Assisted Solution

Objectives

- Match the team's researched problem to the most appropriate build track: chatbot, classifier, dashboard, or awareness campaign.
- Build a working first version using a no-code AI tool.
- Test the prototype with a peer team and make at least one evidence-based iteration.

Timing (90 min)

- 0–10 min: Recap research; introduce the four build tracks with one example each.
- 10–20 min: Teams choose their track, guided by facilitator matching questions.

- 20–70 min: Build sprint — teams work at track-specific stations, facilitator rotates every 8–10 minutes.
- 70–80 min: Peer testing — teams swap prototypes with another team and complete a feedback form.
- 80–90 min: Teams make one concrete iteration and save/export their work.

Step-by-step

1. Recap each team's Problem Brief in one sentence. On the board, sketch the four tracks and one guiding question for each: "Does your problem need someone to answer questions or find something? → Chatbot. Does it need sorting or recognising images? → Classifier. Does it need to show a trend or pattern in numbers? → Dashboard. Does it need behaviour or awareness to change? → Awareness campaign."
2. Each team picks one track (facilitator may gently redirect a mismatched choice, e.g., a "broken taps" team leaning toward a campaign when a chatbot FAQ would serve residents better).
3. Set up four physical or virtual stations, each with the relevant tool open. Give a 3-minute live demo of each track's tool at the front before releasing teams to build (see Build detail for each track).
4. During the 50-minute build sprint, rotate through every team every 8–10 minutes; ask "what problem from your research does this feature solve?" to keep the build grounded in evidence, not novelty.
5. At 70 minutes, pair each team with another team (ideally a different track) for peer testing. Distribute the feedback form: "What worked? What confused you? One suggestion?"
6. Teams spend the final 10 minutes making one visible improvement based on feedback, then save their work: export/share links to the facilitator's email or class Drive folder as backup.

Build detail

Track 1 — Chatbot. Tool: Google AI Studio (aistudio.google.com), free with a Google account.

1. Go to aistudio.google.com and sign in. Click "Create new" → "Chat prompt."
2. In the System Instructions box, paste a persona and knowledge base built from the team's research and interview notes.

3. Set Temperature to 0.3–0.5 in the model settings for a more consistent, factual tone.
4. Test the bot with realistic user messages drawn from the interviews.
5. Export/share: copy the prompt link or screen-record a short demo as backup.

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Example system instruction:
"You are 'Maji Msaidizi' (Water Helper), a friendly chatbot for residents of Kawangware.
You help people find the nearest working water point, report a broken tap, and get simple
water-saving tips. Greet users in English and a little Swahili. If you don't know an exact
location or fact, say so honestly and suggest calling the local Water Board office. Keep
every answer under 60 words and use simple language."

Example test messages:
"Where can I get water near Stage A right now?"
"The tap outside the DC's office has been broken for a week, what should I do?"
"How can my family use less water when supply is unreliable?"
```

Good result: the bot replies in-persona, gives a specific and honest answer or admits uncertainty with a next step, and stays under the word limit. *Iterate:* add 3 more real FAQ scenarios from actual interview answers and refine the system instructions until responses feel locally accurate.

Track 2 — Classifier. Tool: Teachable Machine (teachablemachine.withgoogle.com), free, no login required.

1. Go to teachablemachine.withgoogle.com → Get Started → Image Project → Standard image model.
2. Create classes matching the problem, e.g. "Recyclable," "Organic," "Hazardous" for waste sorting, or "Safe crossing," "Unsafe crossing" for school safety photos.
3. Upload or webcam-capture 15–20 sample images per class with varied angles, lighting and backgrounds (use photos brought from home or taken during the session).
4. Click "Train Model." Test live using the webcam by holding up new sample items or photos.
5. Click "Export Model" to get a shareable link or embed code.

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Example class set for a waste-sorting classifier:
Class 1: "Recyclable" (plastic bottles, cans, paper)
Class 2: "Organic" (banana peels, vegetable scraps)
Class 3: "Hazardous" (batteries, broken glass)

Test procedure: hold up 10 new items not used in training; count how many are
classified correctly; note which class gets confused most often.
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Good result: the model correctly classifies at least 8 out of 10 new test images. *Iterate:* add more training images specifically for the class that is being confused, and retrain.

Track 3 — Dashboard. Tool: Google Forms (data collection) + Google Sheets (charts) + Gemini (analysis).

1. Create a Google Form with questions matching the research issue, e.g. "Rate water availability today, 1–5," or "How many minutes late was your matatu?"
2. Collect at least 10 responses from classmates, family, or Session 1 interview subjects (can include data already gathered).
3. Open the linked Google Sheet, select the response data, then Insert → Chart, choosing a bar or line chart to match the data type.
4. Paste a summary of the data into Gemini or ChatGPT and ask for the two biggest patterns and one recommended action.

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Example analysis prompt (paste the data table above the prompt):
"Here is survey data on matatu delays along Route 46 collected from 12 riders this week.
Identify the 2 biggest patterns in this data and suggest one specific action the route
sacco could take to reduce delays. Keep the answer under 120 words."
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Good result: one to two clear charts plus a written 3-bullet insight summary that a real decision-maker could act on. *Iterate:* if patterns are unclear, collect more responses or add a filter/second chart type.

Track 4 — Awareness Campaign. Tool: Canva (canva.com/education) Magic Media + Gemini/ChatGPT.

1. In Canva, start a new "Instagram Post" or "Poster" design.
2. Open Magic Media (AI image generator) inside Canva and generate a supporting illustration.
3. Use Gemini/ChatGPT to draft slogans and a short video script.
4. Paste the chosen slogan into the Canva design, adjust layout, and export as an image; optionally record the script aloud on a phone.

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Example Magic Media prompt:
"A colourful, flat-design poster illustration of Nairobi schoolchildren safely crossing
a road with a crossing guard, bright morning light, friendly and simple style."

Example copywriting prompt:
"Write 3 short, punchy awareness slogans (under 10 words each) in English and Sheng about
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school road safety for Nairobi students. Then write a 60-second video script aimed at
parents, ending with one clear action they should take."
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Good result: a poster with one clear image and one clear message (not a paragraph of text), plus a short script ready to read aloud. *Iterate:* peer-test the poster with the question "what should I do after seeing this?" — if the answer isn't obvious, cut the text and simplify.

Checks for understanding

- How does your prototype directly respond to something you learned in your research or interviews?
- What did the peer testing round reveal that you didn't expect?
- If you had one more hour, what is the single most important improvement you would make?

Common mistakes & fixes

MISTAKE	FIX
Chatbot gives long, robotic answers	Shorten the system instruction and explicitly cap response length (e.g., "under 60 words")
Classifier trained on too few or too-similar images	Require a minimum of 15 images per class with varied angles and lighting; retrain
Dashboard has too little data to show a real pattern	Pool data across the whole class or clearly label a small sample as illustrative
Awareness poster is cluttered or message is unclear	Apply the "one message, one image" rule and cut text by at least half

Session 3: Present — Pitch the Solution

Objectives

- Deliver a clear, evidence-based 3-minute pitch using a Problem–Solution–Impact–Ask structure.
- Give and receive constructive feedback using a shared rubric.
- Reflect on the design process and receive a Changemaker certificate (and, for one team, the badge).

Timing (90 min, adjust presentation block for number of teams)

- 0–10 min: Welcome guests, explain the pitch format and judging criteria.

- 10–25 min: Final polish — teams rehearse, assign speaking roles, finalise visual aid.
- 25–70 min: Pitches — each team presents for 3 minutes plus 2 minutes Q&A (roughly 5 min per team; for 6 teams this is 30 min, extend or trim rehearsal time accordingly).
- 70–80 min: Judges tally scores while students and parents mingle over snacks.
- 80–88 min: Awards ceremony — Changemaker badge announced, all certificates distributed.
- 88–90 min: Closing reflection circle and group photo.

Step-by-step

1. Set up the room: judging table at the front, audience seating, projector or table space for team posters/prototypes.
2. Explain the pitch structure to the whole room: Hook (10 seconds — a real story from research), Problem (the evidence, including any stats verified against a human source), Solution (a live or recorded demo of the prototype), Impact (who benefits and how you'd know it's working), Ask (what's needed next — funding, a partner organisation, adoption by a school or sacco).
3. Hand out the pitch template and give teams 15 minutes to rehearse with a timer; give each team one specific piece of feedback during rehearsal.
4. Explain Q&A norms to the audience: one question per person, phrased to help the team improve, no repeat questions.
5. Run the pitches in order, keeping strict time with the visible countdown; judges score using the rubric sheet as each team finishes.
6. While judges tally scores, invite parents to try prototypes hands-on at each team's table.
7. Announce the Changemaker badge winner, present all Changemaker certificates, and take a group photo.
8. Close with a one-sentence reflection round: "One thing I'd do differently next time is..."

Build detail

No new AI build tool is introduced this session; the focus is polishing communication using tools from Sessions 1–2.

1. In Canva, teams build a simple 3–5 slide pitch deck using the provided template (Title, Problem + evidence, Solution demo screenshot, Impact, Ask).
2. Teams paste their draft pitch script into Gemini/ChatGPT and ask for a tightened version.
3. Teams rehearse aloud, timing themselves, and prepare a backup screenshot or short screen-recording of their prototype in case of a live demo failure.

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Example script-tightening prompt:
"Here is my 3-minute pitch script about school road safety in Nairobi. Make it punchier,
cut filler words, keep it under 400 words total, and suggest one strong closing line that
makes the audience want to act."
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Good result: a script under 400 words that a nervous 13-year-old can deliver in under 3 minutes, with a clear ask at the end. *Test/iterate:* time the rehearsal out loud twice; cut content (not speaking speed) if over time.

Checks for understanding

- What is the difference between showing a feature of your prototype and showing its impact?
- How would you explain to a judge why you chose this AI tool and not one of the other three tracks?
- What is one thing feedback from rehearsal or Q&A taught you that you'll apply next time?

Common mistakes & fixes

MISTAKE	FIX
Team reads directly off notes or screen	Rehearse with notecards limited to 5 keywords per card, not full sentences
Pitch skips the problem/evidence and jumps to the solution	Check every pitch against the Problem–Solution–Impact–Ask checklist before the final rehearsal
Live demo fails on stage	Require every team to prepare a backup screenshot or short screen-recording in advance
Pitch runs over time	Use a visible countdown during rehearsal; trim content, not delivery speed

Assessment & Showcase

Assessment is formative and cumulative across all three sessions rather than a single test. The facilitator keeps a running checklist per team covering: quality and specificity of the Session 1 Problem Brief (including at least one AI claim flagged and verified), functionality of the Session 2 prototype (does it actually run and respond to test inputs), evidence of iteration after peer feedback, and clarity and evidence-base of the Session 3 pitch.

The Session 3 showcase is scored by a panel of 2–3 judges (the facilitator plus one or two invited parents or a guest mentor) using a simple rubric with four equally weighted categories: Research depth & evidence (25%), Technical execution / AI tool use (25%), Creativity & community relevance (25%), Pitch clarity & delivery (25%). Scores are tallied immediately after presentations while parents interact with the prototypes. The highest-scoring team receives the "TinkerWith_ Changemaker" badge; every student, regardless of placing, receives a Changemaker certificate.

"Done" for each team means: a completed one-page Problem Brief with map, a working (even if simple) prototype with a saved/shared link or exported screenshots, a delivered 3-minute pitch with Q&A, and a compiled portfolio (physical folder or shared Drive folder) containing all three artefacts plus a one-paragraph reflection on what the team would do next with more time.

Extension & Home Challenges

- **Easy:** Take your prototype home and test it with 5 family members or neighbours. Record their reactions and write 3 specific improvement notes based on their feedback.
- **Medium:** Extend your prototype with one new dimension — add a Swahili or Sheng response set to your chatbot, add a second class to your classifier, add a second data question to your dashboard, or produce a second version of your campaign for a different audience (e.g., parents vs. drivers). Retest and compare accuracy or reception before and after.
- **Hard:** Contact a real local organisation connected to your challenge (a school administration, a water committee, a matatu sacco, or a county waste office). Share your prototype with them, collect their genuine feedback, and write a

one-page adoption proposal describing what would need to change for them to actually use your solution. Optionally, build a second track (e.g., pair your classifier with a dashboard showing sorting results over time) to create a combined, more powerful solution.